

Mutualism

Today's Agenda:

- Quiz
- Preparing for the midterm
 - Network assignment
 - Mutualism

Preparing for the midterm

- Next time is the midterm
- Most of the questions will be recycled from the quizzes
 - May be modified (e.g., changed to multiple choice, reworded, etc)
 - So look over quizzes, particularly missed questions!
- May be a few new things pulled from lectures or the book

Network Assignment

Consists of 3 parts:

Network Assignment

Consists of 3 parts:

- 1) Read papers before class on March 3!
 - a) 16 options given (see the assignment details on Canvas for the list)
 - b) Don't have to read all, but should read a subset of them
 - c) **In-class discussion will be part of your grade!**

Network Assignment

Consists of 3 parts:

- 1) Read papers before class on March 3!
- 2) Write about network(s) that might be useful for your question/system
 - a) Explain which network type(s) or representation(s) seem best suited to your system/question
 - b) Explain **why** these types/representations are well suited
 - c) Explain why other types/representations may **not be** a good fit.
 - d) Support your arguments with ideas from class, your readings, or your own reasoning, but remember to cite anything that comes from a published article.
 - e) One to two pages, double spaced (citations don't count against the word limit)

Network Assignment

Consists of 3 parts:

- 1) Read papers before class on March 3!
- 2) Write about network(s) that might be useful for your question/system
- 3) How might network ecology help us to address big questions in ecology?
 - a) E.g., networks are integrated into species distribution modelling via joint SDMs, allowing prey distributions to inform predator distributions.
 - b) Write a short explanation (a page or so, double spaced) about how a major topic in ecology might benefit from integrating ecological networks.
 - c) For potential ideas of “major topics” you can look to your syllabus (competition, coexistence, biodiversity-ecosystem functioning, etc.), but aren’t limited to what is mentioned there.

Network Assignment

Grade Breakdown:

- In-class participation and discussion 30%
- Written portion 1 50%
- Written portion 2 20%

Network Assignment

Questions?

Definitions



OUR CURRENT UNDERSTANDING OF MUTUALISM

JUDITH L. BRONSTEIN

*Department of Ecology and Evolutionary Biology, University of Arizona
Tucson, Arizona 85721 USA*

Species A

	+	0	-
+	?	?	?
0	?	?	?
-	?	?	?

Definitions



OUR CURRENT UNDERSTANDING OF MUTUALISM

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Species A

	+	0	-
+	Mutualism	Commensalism	Predation Parasitism
0	Commensalism	Neutralism	Amensalism
-	Predation Parasitism	Amensalism	Competition

Species B

Definitions

Facilitation: one species alters the environment in such a way that it benefits another species

Definitions

Obligate vs facultative mutualisms

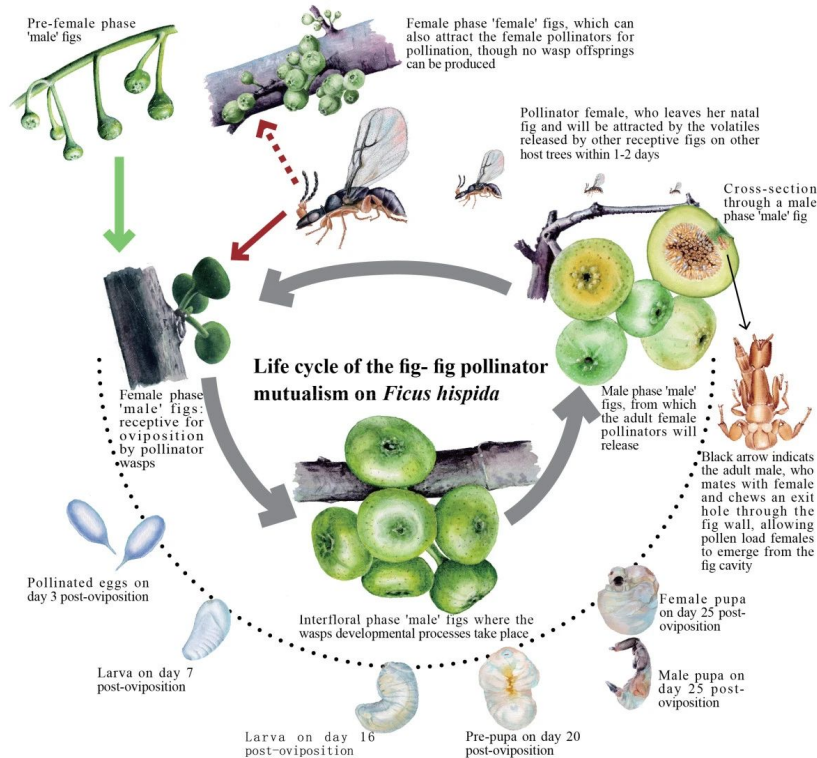
- Obligate: need a specific partner to live/reproduce/etc.
- Facultative: don't require a specific partner

Definitions

Obligate vs facultative mutualisms

- Obligate: need a specific partner to live/reproduce/etc.
- Facultative: don't require a specific partner
- Just because one partner is an obligate doesn't mean the other is
- A lot of (most?) interactions are asymmetrical

Obligate Mutualism Example



Obligate mutualism within a host drives the extreme specialization of a fig wasp genome

Jin-Hua Xiao^{1,2}, Zhen Yue³, Ling-Yi Jia^{1,3,4}, Xin-Hua Yang^{1,2}, Li-Hua Niu^{1,2}, Zhao Wang^{1,2}, Peng Zhang^{1,2}, Bao-Fa Sun^{1,2}, Shun-Min He^{1,2}, Zi Li^{1,2}, Tian-Lin Xiong^{1,2}, Wen Xin¹, Hai-Feng Gu^{1,2}, Bo Wang^{1,2}, John H. Werren⁵, Robert W. Murphy^{6,7}, David Wheeler⁸, Li-Ming Niu⁹, Guang-Chang Ma¹, Ting Tang^{1,2}, Sheng-Nan Ban¹, Ning-Xin Wang¹, Chun-Yan Yang¹, Nan Wang¹, Yue-Guan Fu¹, Wen-Zhu Li¹, Soojin V. Yi^{1,2}, Xing-Yu Yang^{1,2}, Qing Zhou¹, Chang-Xin Lu¹, Chun-Yan Xu¹, Li-Juan He¹, Li-Li Yu¹, Ming Chen¹, Yuan Zheng¹, Shao-Wei Wang¹, Shuang Zhao¹, Yan-Hong Li¹, Yang-Yang Yu¹, Xiao-Ju Qian¹, Yue Cai¹, Lian-Le Ban¹, Shu Zhang¹, Jun-Yi Wang¹, Ye-Ye¹, Hua Xiao¹, Guan-Hong Wang¹, Hua Yu¹, Wen-Shan Wu¹, James M. Cook^{10,11}, Jun Wang¹ and Di-Wei Huang^{10,11}

Facultative Mutualism Example



Cylindropuntia imbricata



Crematogaster opuntiae



Liometopum apiculatum



Oikos 116: 500–512, 2007
doi: 10.1111/j.2007.0030-1299.15317.x,
Copyright © Oikos 2007, ISSN 0030-1299
Subject Editor: Jane Memmott, Accepted 10 November 2006

Does having multiple partners weaken the benefits of facultative mutualism? A test with cacti and cactus-tending ants

Tom E. X. Miller

T. E. X. Miller (tmiller2@unl.edu), School of Biological Sciences, Univ. of Nebraska – Lincoln, 348 Manter Hall, Univ. of Nebraska, Lincoln, NE 68588-0118, USA.

Facultative Mutualism Example



Cylindropuntia imbricata
Extrafloral nectaries



Crematogaster opuntiae
Bad defense



Liometopum apiculatum
Good Defense



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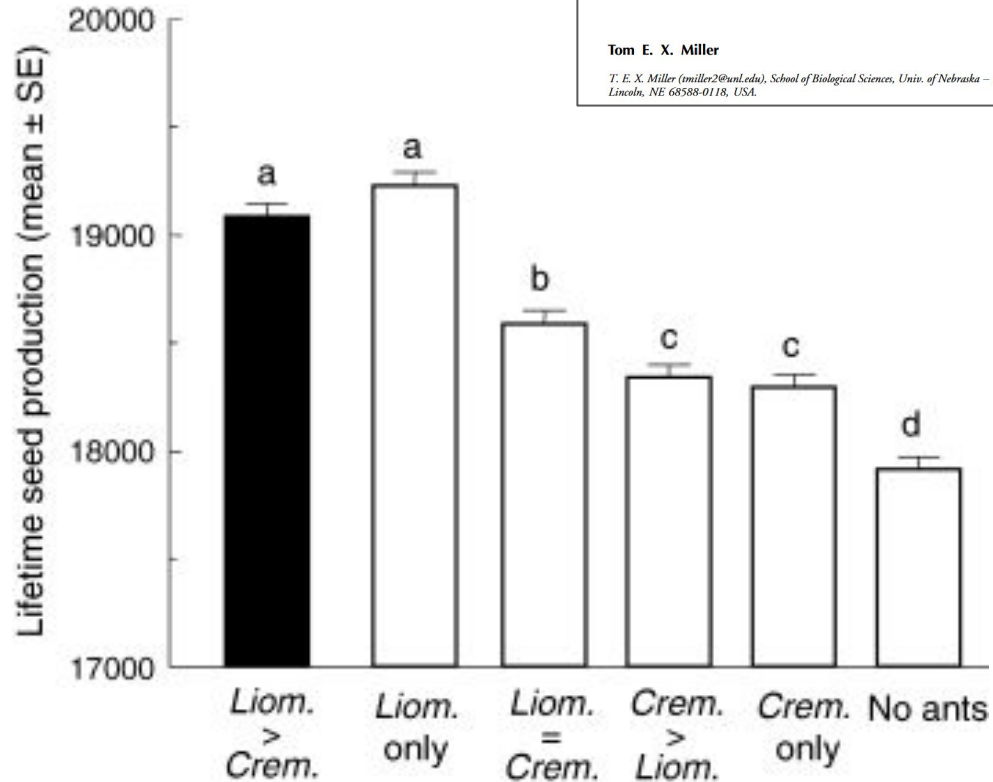
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Facultative Mutualism Example



Cylindropuntia imbricata



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Facultative Mutualism Example

Species A



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Species B

	+	0	-
+	 Mutualism	 Commensalism	Predation Parasitism
0	Commensalism	Neutralism	Amensalism
-	Predation Parasitism	Amensalism	Competition

Evolution of Mutualism

Costs vs benefits

- Evolution only 'cares' about net benefits to a species
- Want to minimize costs
- Selection pushes towards parasitism

		Species A		
		+	0	-
Species B	+	Mutualism	Commensalism	Predation Parasitism
	0	Commensalism	Neutralism	Amensalism
	-	Predation Parasitism	Amensalism	Competition

Evolution of Mutualism

How are mutualisms maintained?

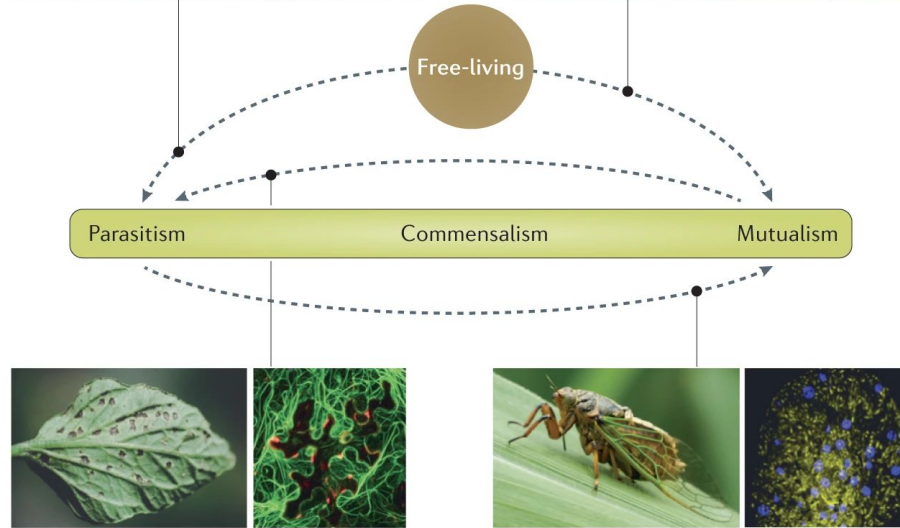
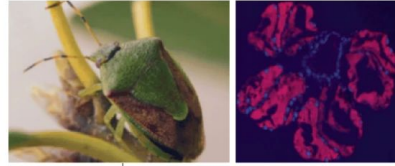
- Partner fidelity (e.g., lichen, chloroplasts)
- Partner choice (only choose reliable partners)

Changes in Evolutionary Time

a Obligate pathogen from environmental ancestor



b Growth benefit from soil-associated bacteria



c HGT of virulence-associated loci

d Replacement of existing nutritional symbiont

Microbial evolution and transitions along the parasite–mutualist continuum

Georgia C. Drew^{1,2}, Emily J. Stevens^{1,2} and Kayla C. King¹

Early issues with integrating mutualism into theory

Why does mutualism lag behind competition and predation?

Early issues with integrating mutualism into theory

Remember these competition models?

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

dN_1/dT = rate of change in N_1
 k_1 = carrying capacity of species 1
 N_1 = # species 1
 N_2 = # species 2
 α_{12} = relative impact of sp. 2 on sp. 1

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

dN_2/dT = rate of change in N_2
 k_2 = carrying capacity of species 2
 α_{21} = relative impact of sp. 1 on sp. 2

Early issues with integrating mutualism into theory

Remember these competition models? - Now they're mutualism models

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 + \alpha_{12} N_2}{k_1} \right)$$

dN_1/dT = rate of change in N_1
 k_1 = carrying capacity of species 1
 N_1 = # species 1
 N_2 = # species 2
 α_{12} = relative impact of sp. 2 on sp. 1

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 + \alpha_{21} N_1}{k_2} \right)$$

dN_2/dT = rate of change in N_2
 k_2 = carrying capacity of species 2
 α_{21} = relative impact of sp. 1 on sp. 2

Early issues with integrating mutualism into theory

Where can species 1 invade?

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 + \alpha_{12} N_2}{k_1} \right)$$

dN_1/dT = rate of change in N_1
 k_1 = carrying capacity of species 1
 N_1 = # species 1
 N_2 = # species 2
 α_{12} = relative impact of sp. 2 on sp. 1

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 + \alpha_{21} N_1}{k_2} \right)$$

dN_2/dT = rate of change in N_2
 k_2 = carrying capacity of species 2
 α_{21} = relative impact of sp. 1 on sp. 2

Zero Net Growth Isoclines (ZNGIs)

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 + \alpha_{12} N_2}{k_1} \right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 + \alpha_{21} N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = r_1 N_1 \left(\frac{k_1 - N_1 + \alpha_{12} N_2}{k_1} \right)$$

$$0 = r_2 N_2 \left(\frac{k_2 - N_2 + \alpha_{21} N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = \left(\frac{k_1 - N_1 + \alpha_{12}N_2}{k_1} \right)$$

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Zero Net Growth Isoclines (ZNGIs)

$$0 = k_1 - N_1 + \alpha_{12}N_2$$

$$0 = k_2 - N_2 + \alpha_{21}N_1$$

Zero Net Growth Isoclines (ZNGIs)

$$N_1 = k_1 + \alpha_{12} N_2$$

$$\alpha_{21} N_1 = N_2 - k_2$$

Zero Net Growth Isoclines (ZNGIs)

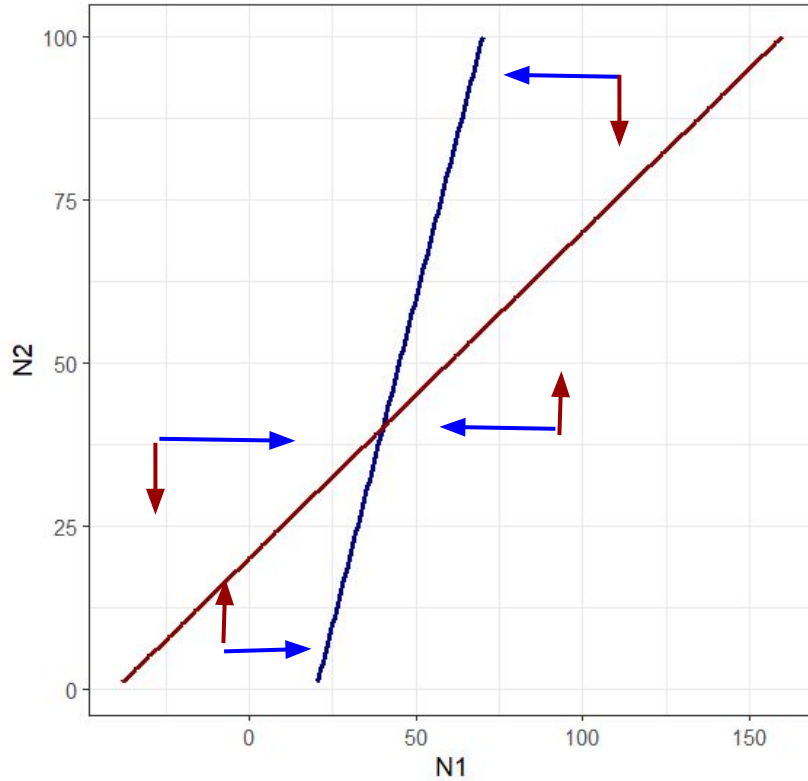
$$N_1 = k_1 + \alpha_{12} N_2$$

$$N_1 = \frac{N_2 - k_2}{\alpha_{21}}$$

Zero Net Growth Isoclines (ZNGIs)

Species 1

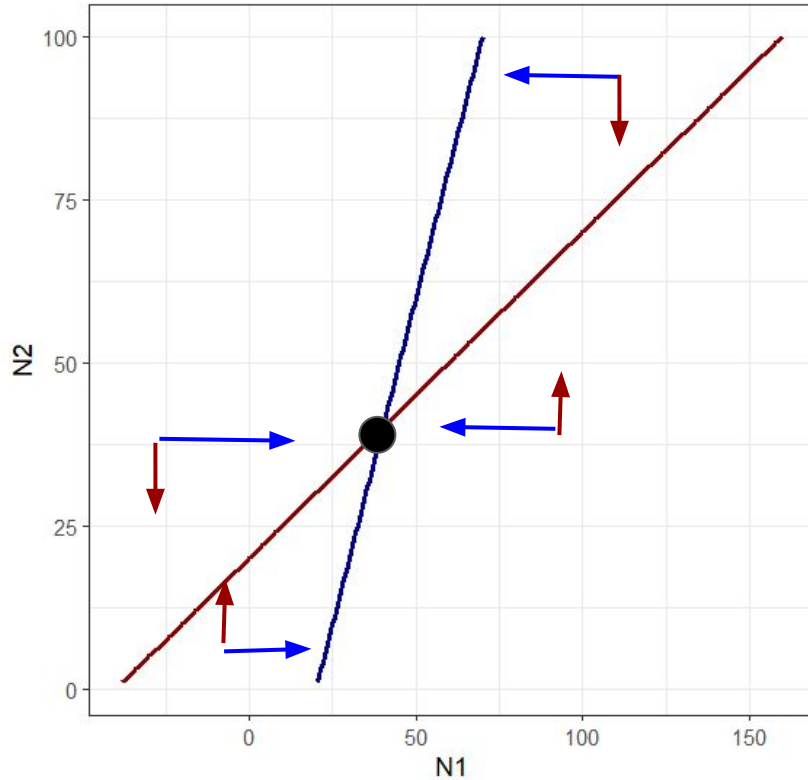
Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1

Species 2

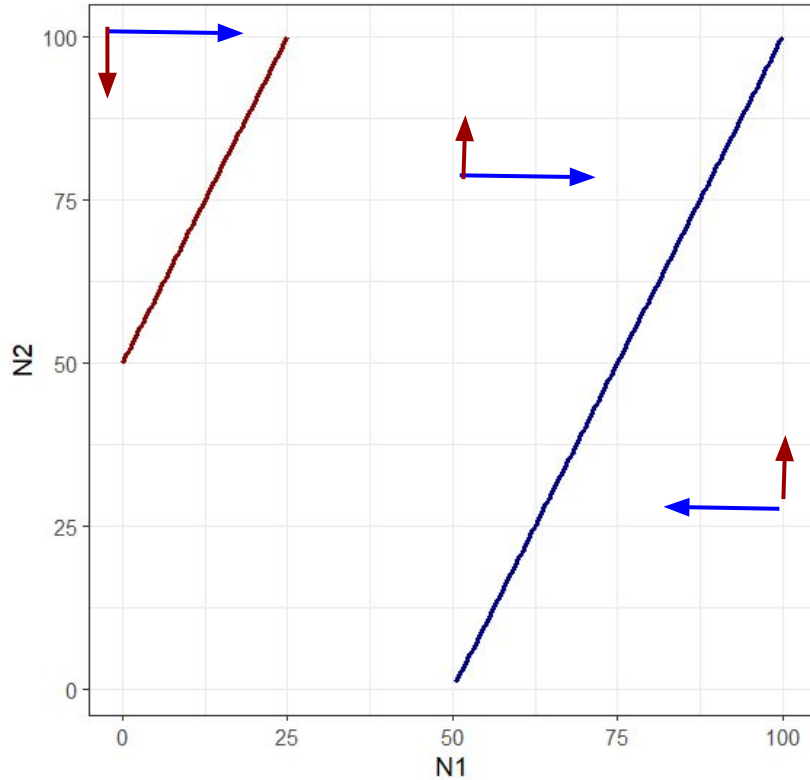


Can lead to coexistence

Zero Net Growth Isoclines (ZNGIs)

Species 1

Species 2



Can also lead to unlimited growth!

Early issues with integrating mutualism into theory

Why don't we see this unlimited growth among mutualistic partners in nature?

Early issues with integrating mutualism into theory

Why don't we see this unlimited growth among mutualistic partners in nature?

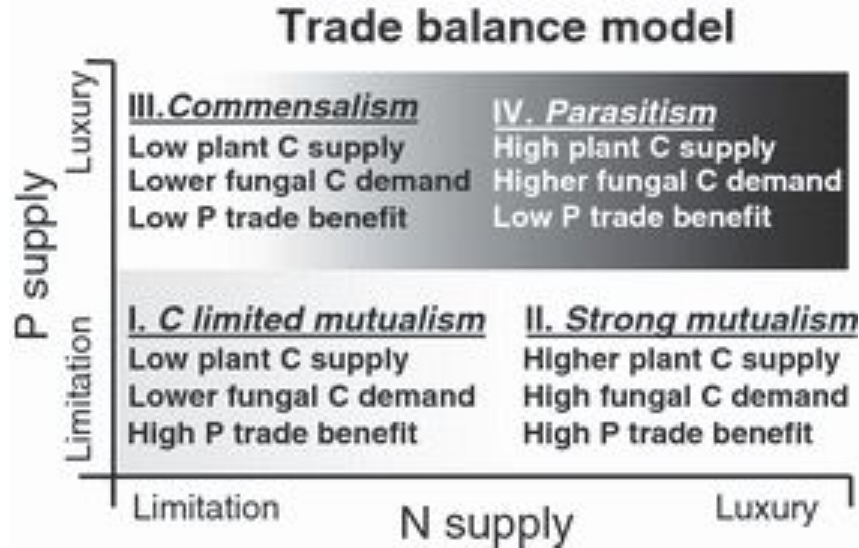
- Mutualisms have costs!
- This is also why they evolution and context can alter them!

Context dependency

Changing conditions may lead to changes in interaction types

		Species A		
		+	0	-
Species B	+	Mutualism	Commensalism	Predation Parasitism
	0	Commensalism	Neutralism	Amensalism
	-	Predation Parasitism	Amensalism	Competition

Context dependency: AMF+Plants





Tansley review

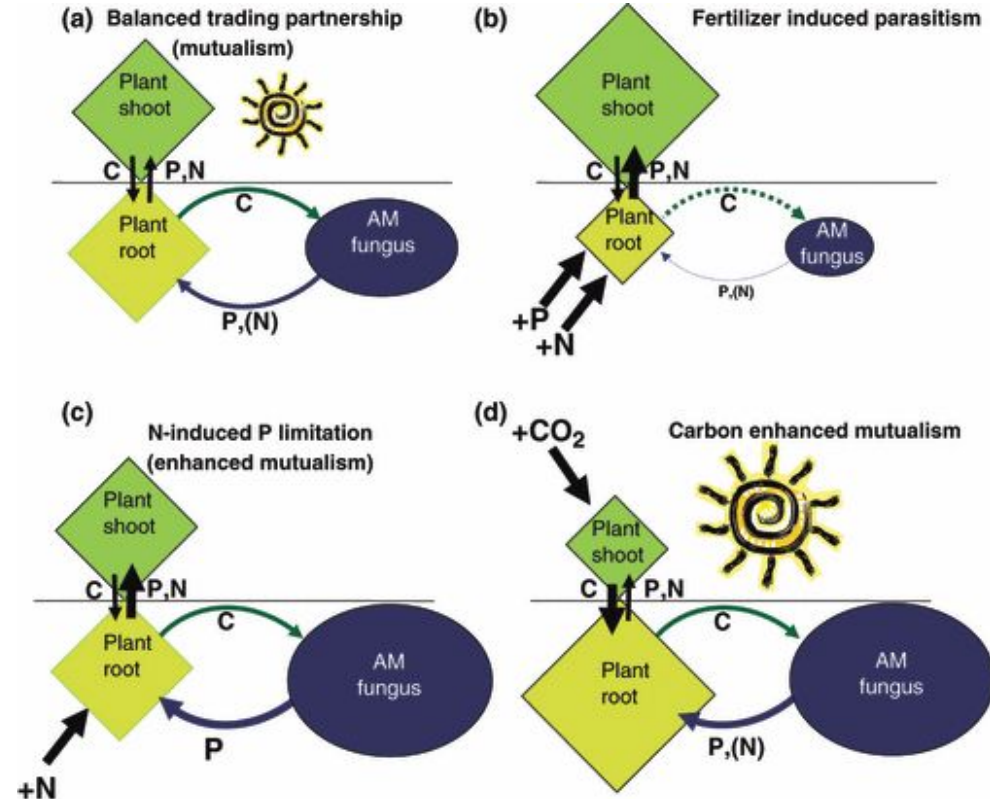
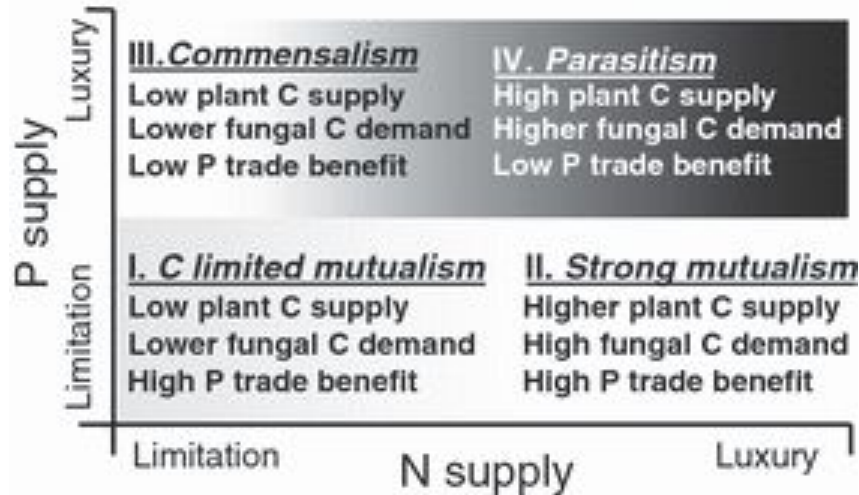
Resource stoichiometry elucidates the structure and function of arbuscular mycorrhizas across scales

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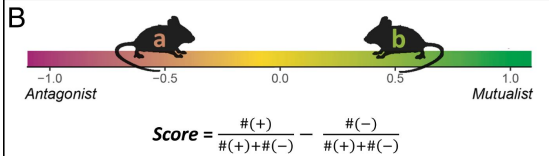
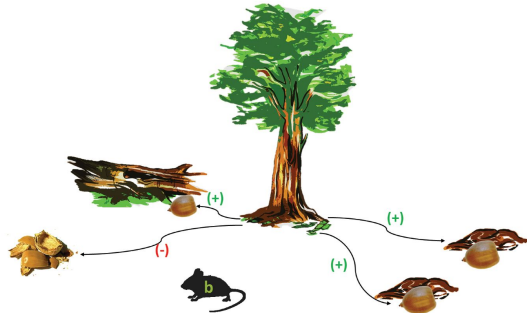
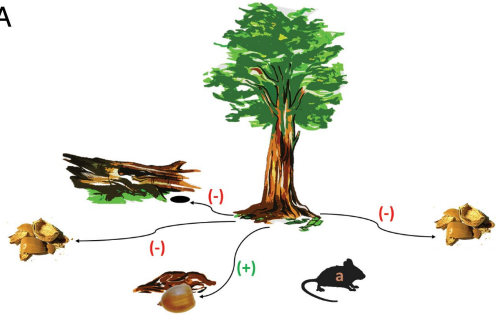
Context dependency: AMF+Plants

Trade balance model

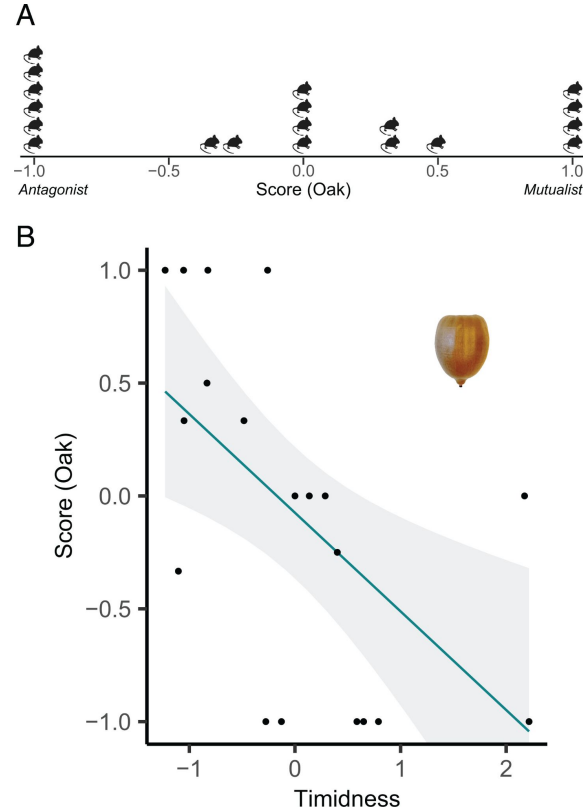
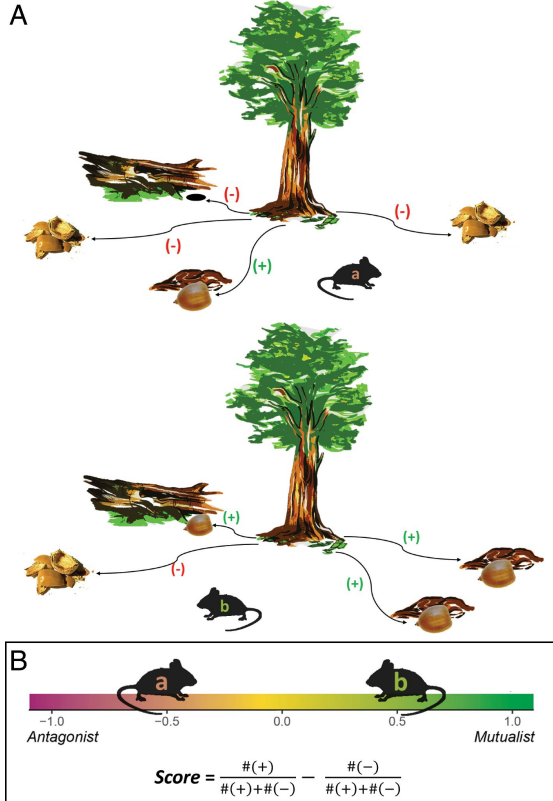


Context dependency: Seeds and Mice

A



Context dependency: Seeds and Mice

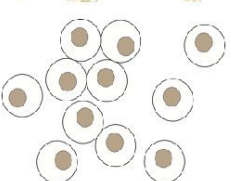
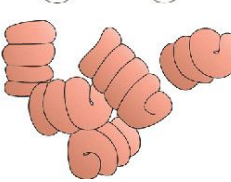


Context dependency: Daphnia parasites

VOL. 145, NO. 5 THE AMERICAN NATURALIST NOVEMBER 2011

Context-Dependent Host-Symbiont Interactions: Shifts along the Parasitism-Mutualism Continuum

Mary A. Rogalski,^{1,2*} Tara Stewart Merrill,^{1,2} Camden D. Gowder,² Carlo E. Cáceres,² and Meghan A. Duffy¹

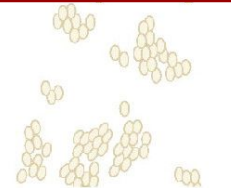
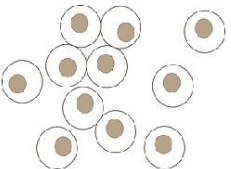
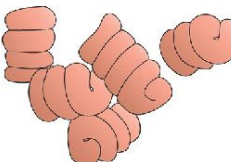
Symbiont	Symbiont cartoon used in figures	Taxon	Site of infection	Impact on host reproduction	Impact on host lifespan	Relevant references
<i>Metschnikowia bicuspidata</i>		fungus	Infects by piercing gut, develops in host hemolymph	Reduces host fecundity (by ~25-50%)	Strong	Duffy and Hall 2008; Auld et al. 2012; Stewart Merrill and Cáceres 2018
MicG (species identity not yet determined)		microsporidian	Gut epithelium	Evaluated in this study	None detected (this study; supplemental text S1, Figure S1)	Genbank accession MH635259
<i>Pasteuria ramosa</i>		bacterium	Infects via host gut (attachment to esophagus); develops in host hemolymph	Very strong (infected hosts rarely reproduce)	No significant impact (except via increased risk of fish predation)	Auld et al. 2012; Duffy et al. 2019
<i>Spirobacillus cienkowskii</i>		bacterium	Hypothesized to infect via host gut; develops in host hemolymph	Very strong (infected hosts rarely reproduce)	Very strong	Duffy & Hall 2008, Wale et al. 2019

Context dependency: Daphnia parasites

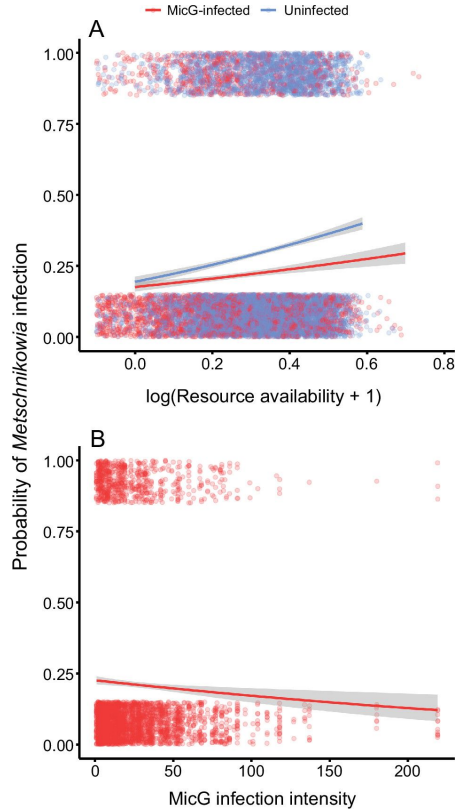
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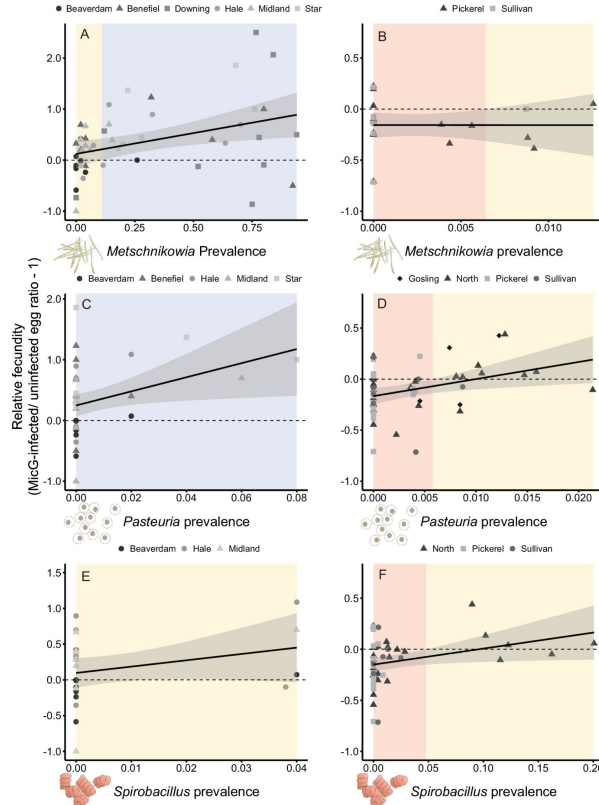
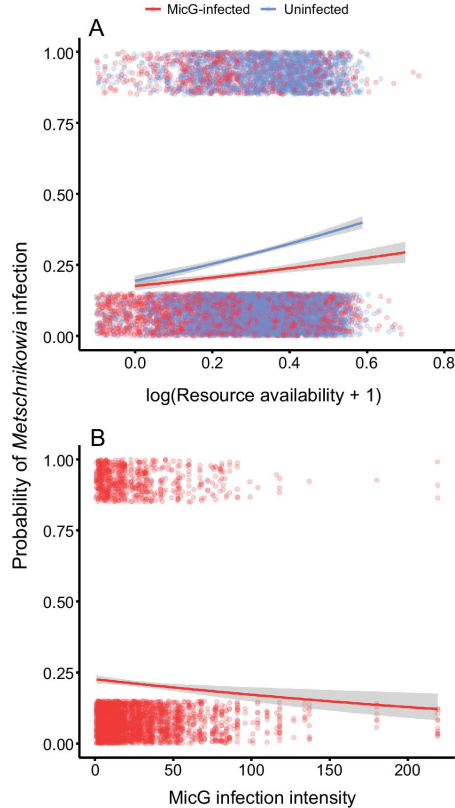


VOL. 198, NO. 5 THE AMERICAN NATURALIST NOVEMBER 2021

Context-Dependent Host-Symbiont Interactions: Shifts along the Parasitism-Mutualism Continuum

Mary A. Rogalski,^{1,2,*} Tara Stewart Merrill,^{3,4} Camden D. Gowler,² Carla E. Cáceres,³ and Meghan A. Duffy²

Context dependency: Daphnia parasites



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Parasitism
Commensalism
Mutualism

Context dependency

		Species A		
		+	0	-
Species B	+	Mutualism	Commensalism	Predation Parasitism
	0	Commensalism	Neutralism	Amensalism
	-	Predation Parasitism	Amensalism	Competition

Context dependency

		Species A		
		+	0	-
Species B	+	Mutualism	Commensalism	Predation Parasitism
	0	Commensalism	Neutralism	Amensalism
	-	Predation Parasitism	Amensalism	Competition

Mutualism and conservation

ECOLOGY LETTERS

Ecology Letters, (2010) 13: 1459–1474 doi: 10.1111/j.1461-0248.2010.01538.x

IDEA AND PERSPECTIVE

Mutualisms in a changing world: an evolutionary perspective

E. Toby Kiers,^{1*} Todd M. Palmer,² Anthony R. Ives,³ John F. Bruno⁴ and Judith L. Bronstein⁵

Abstract

There is growing concern that rapid environmental degradation threatens mutualistic interactions. Because mutualisms can bind species to a common fate, mutualism breakdown has the potential to expand and accelerate effects of global change on

Driver	Mutualism type	Ecological and evolutionary responses	Critical conservation questions
Climate change	Coral–zooxanthellae mutualisms	Expulsion and/or switching of mutualist partners ^{1,3}	Does switching facilitate invasion by antagonistic partners or increase host resilience?
	Plant–pollinator mutualisms	Changes in phenologies decouple plant and pollinator partners ^{4,7}	How resilient are hosts to changes in partner densities?
Nutrient enrichment	Plant–rhizosphere mutualisms	Spread of less effective but highly competitive mutualist partners ^{8,10}	Can evolutionary shifts towards antagonism be reversed?
Habitat fragmentation	Plant–pollinator mutualisms	Reductions in pollinator abundance and richness ^{11,12} ; changes in recruitment, gene flow and selection regimes for mutualistic partners ^{13,14}	Will ‘rescue’ partner species effectively replace native partners?
Overhunting	Plant–disperser mutualisms	Changes in tree recruitment patterns leading to shifts in forest composition ¹⁵ ; reduction of long-distance dispersal events crucial for maintaining genetic structure of tree populations ¹⁶	Can landscape-level effects be predicted?
Exotic species introductions	Plant–rhizosphere mutualisms	Contrasting effects: degradation of native rhizosphere communities ¹⁷ vs. enhancement of exotics by native mutualists ^{18,19}	Can evolutionary shifts towards antagonism be reversed?
	Ant–plant mutualisms	Changes in seed dispersal and pollination patterns upon exotic ant invasion ^{20–21}	How rapidly can hosts respond to changes in partner species?
	Coral–zooxanthellae mutualisms	Naturalization of exotic symbiont into coral ecosystem with unknown evolutionary consequences ²²	Do exotic mutualistic partners displace natives? How will displacement affect hosts?
Loss of third species	Ant–plant mutualisms	Shifts in cost:benefit ratios of a mutualism via loss of an external species (e.g., herbivores) ²³	What is the role of external species in directing mutualistic outcomes between two partners?

Each example illustrates the conservation question associated with the mutualism listed, but questions apply broadly to a wide range of threatened mutualisms not listed here.

References for footnotes are listed in main text and/or Appendix S1 references: ¹Jones *et al.* (2008), ²LaJeunesse *et al.* (2009), ³Stat *et al.* (2008), ⁴Yang & Rudolf (2010), ⁵Hegland *et al.* (2009), ⁶Doi *et al.* (2008), ⁷Memmott *et al.* (2007), ⁸Egerton-Warburton *et al.* (2007), ⁹Nijjer *et al.* (2010), ¹⁰Johnson (2010), ¹¹Winfree *et al.* (2009), ¹²Potts *et al.* (2010), ¹³Murua *et al.* (2010), ¹⁴Eckert *et al.* (2009), ¹⁵Terborgh *et al.* (2008), ¹⁶Jordano *et al.* (2007), ¹⁷Wolfe *et al.* (2008), ¹⁸Reinhart & Callaway (2006), ¹⁹Carey *et al.* (2004), ²⁰Rodriguez-Cabal *et al.* (2009), ²¹Lach (2008), ²²LaJeunesse *et al.* (2005) and ²³Palmer *et al.* (2008).

Next class: Midterm